Class 18 : Investigating Pertussis Resurgence: Mini Project

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Background

Pertussis (a.k.a whooping cough) is a common lung infectuion caused by the bacteria B.Pertussis.

This infection can affect people all ages but is most severe for those under 1 year of age.

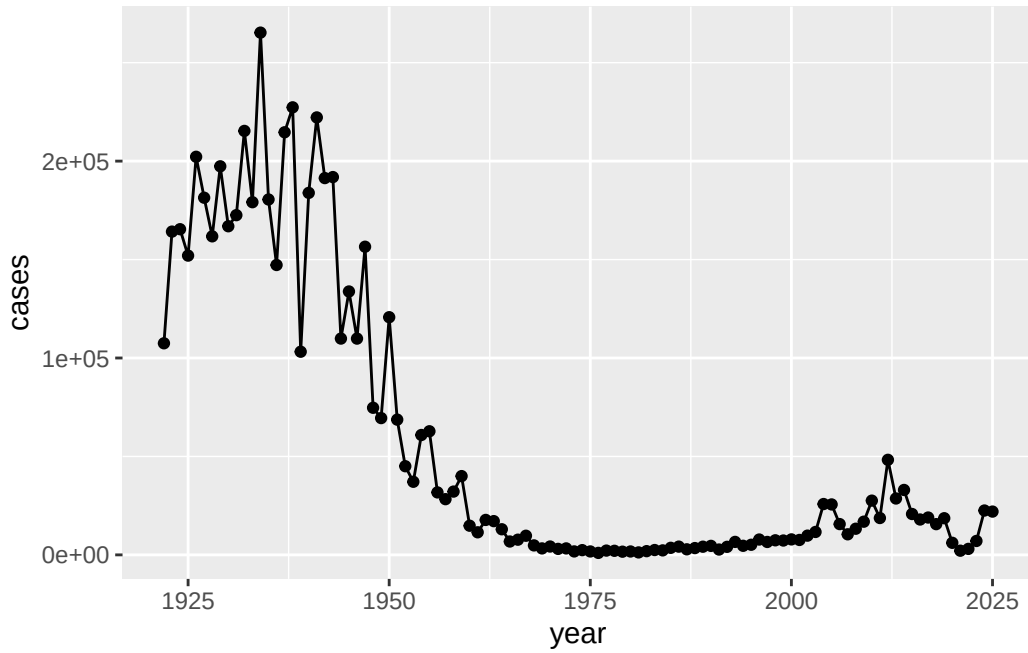
The CDC track the number of reported cases in the US: We can "scrape" this data with the **datapasta** package.

```
cdc <- data.frame(
                                 year = c(1922L,1923L,1924L,1925L,
                                          1926L,1927L,1928L,1929L,1930L,1931L,
                                          1932L,1933L,1934L,1935L,1936L,
                                          1937L,1938L,1939L,1940L,1941L,1942L,
                                          1943L,1944L,1945L,1946L,1947L,
                                          1948L,1949L,1950L,1951L,1952L,
                                          1953L,1954L,1955L,1956L,1957L,1958L,
                                          1959L,1960L,1961L,1962L,1963L,
                                          1964L,1965L,1966L,1967L,1968L,1969L,
                                          1970L,1971L,1972L,1973L,1974L,
                                          1975L,1976L,1977L,1978L,1979L,1980L,
                                          1981L,1982L,1983L,1984L,1985L,
```

```
                                        1986L,1987L,1988L,1989L,1990L,
                                        1991L,1992L,1993L,1994L,1995L,1996L,
                                        1997L,1998L,1999L,2000L,2001L,
                                        2002L,2003L,2004L,2005L,2006L,2007L,
                                        2008L,2009L,2010L,2011L,2012L,
                                        2013L,2014L,2015L,2016L,2017L,2018L,
                                        2019L,2020L,2021L,2022L,2023L,2024L,2025L),
                                cases = c(107473,164191,165418,152003,
                                        202210,181411,161799,197371,
                                        166914,172559,215343,179135,265269,
                                        180518,147237,214652,227319,103188,
                                        183866,222202,191383,191890,109873,
                                        133792,109860,156517,74715,69479,
                                        120718,68687,45030,37129,60886,
                                        62786,31732,28295,32148,40005,
                                        14809,11468,17749,17135,13005,6799,
                                        7717,9718,4810,3285,4249,3036,
                                        3287,1759,2402,1738,1010,2177,2063,
                                        1623,1730,1248,1895,2463,2276,
                                        3589,4195,2823,3450,4157,4570,
                                        2719,4083,6586,4617,5137,7796,6564,
                                        7405,7298,7867,7580,9771,11647,
                                        25827,25616,15632,10454,13278,
                                        16858,27550,18719,48277,28639,32971,
                                        20762,17972,18975,15609,18617,
                                        6124,2116,3044,7063,22538,21996)
      )
```

Q. Make a plot of `year` vs `cases`

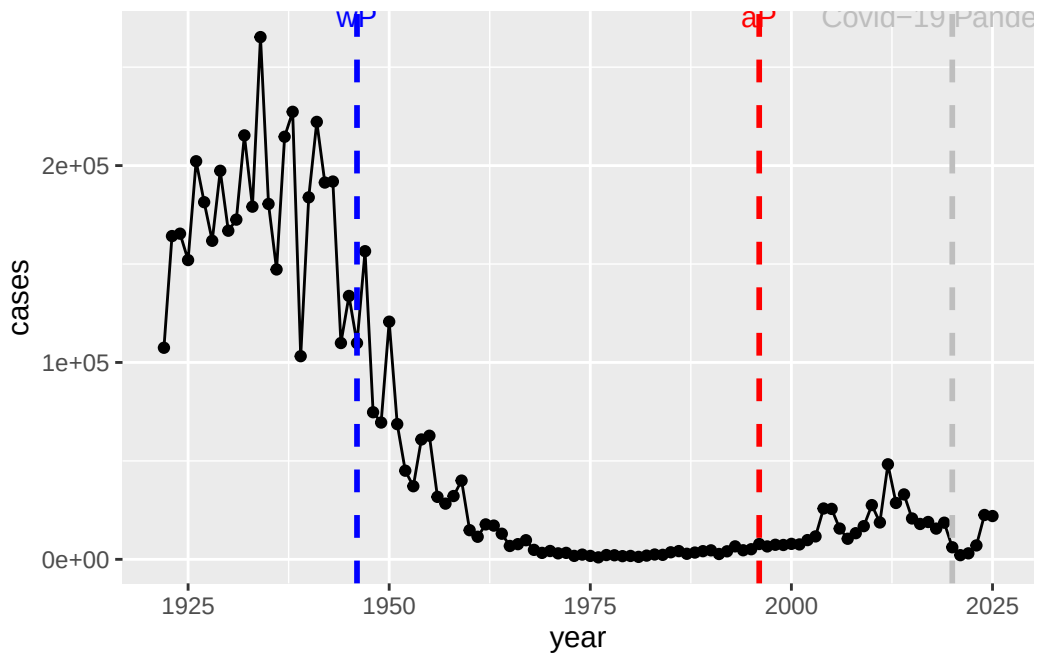
```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point()+
  geom_line()+
  labs()
```



Q. Add some major milestones including the first wP vaccine roll-out (1946), and the switch to the newer aP vacccine (1996).

```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point()+
  geom_line()+
  labs()+
  geom_vline(xintercept = 1946, color = "blue", linewidth = 1, linetype = "dashed")+
  geom_vline(xintercept = 1996, color = "red", linewidth = 1, linetype = "dashed")+
  geom_vline(xintercept = 2020, color = "gray", linewidth = 1, linetype = "dashed")+
  annotate("text",
           x = 1946,
           y = max(cdc$cases),
           label = "wP",
           color = "blue",
           vjust = -0.5)+
  annotate("text",
           x = 1996,
           y = max(cdc$cases),
           label = "aP",
           color = "red",
```

```
          vjust = -0.5)+
annotate("text",
          x = 2020,
          y = max(cdc$cases),
          label = "Covid-19 Pandemic",
          color = "gray",
          vjust = -0.5)
```



The graph shows were high case numbers in the pre 1940s, they were reduced drastically based on the rolling out of the wP vaccine. The numbers then began to rise again in the 1980s due to the fact that documentaries and studies came out that falsely linked other sicknesses to vaccines. This lead to less vaccines being given and a rise in infections. Another correlation that can be made from the graph below and requires further research is the fact that the aP induced protection wanes faster than wP but why? and does this cause the rise of infection as well?

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB project

Computational Models of Immunity - pertussis boost project aims to provide the scientific community with this very information.

They make their data available via JSON format returning API. We can read this in R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)

subject <- read_json("http://cmi-pb.org/api/v5_1/subject", simplifyVector=TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. How many wP and aP individuals are in this `subject` table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the biological sex breakdown?

```
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. In terms of race and gender is this dataset representative of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more database tables:

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge) the different tables so we have all the data in one place not spread accross different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
```

```
2                            1                            1         Blood
3                            3                            3         Blood
4                            7                            7         Blood
5                           11                           14         Blood
6                           32                           30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
```

```
1                           -3                              0         Blood
2                           -3                              0         Blood
3                           -3                              0         Blood
4                           -3                              0         Blood
5                           -3                              0         Blood
6                           -3                              0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q. What antibody isotypes are measure for these patients?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

Q. How many datasets are there in the abdata?

```r
unique(abdata$dataset)
```

```
[1] "2020_dataset" "2021_dataset" "2022_dataset" "2023_dataset"
```

Q. How many times do the datasets appear in the abdata? what do you notice from this?

```r
table (abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

We can see 4 different data sets and from them we can see that there were so many data sets in 2020 likely the result of the covid pandemic while there was an extremely low amount in the 2021 and 2022 data sets with an increase again in 2023. Although something to notice is tat even in 2023 we dont have half of the entries that we did from 2020.

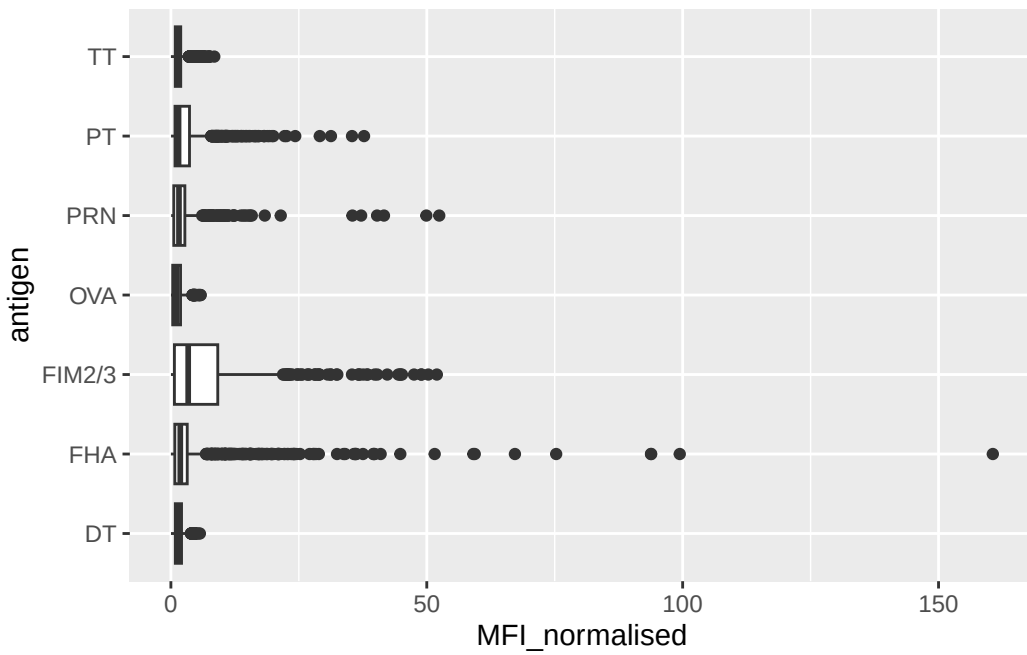
Let's focus on the IgG isotype and make a plot of MFI_normalized for all antigens.

```r
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
```

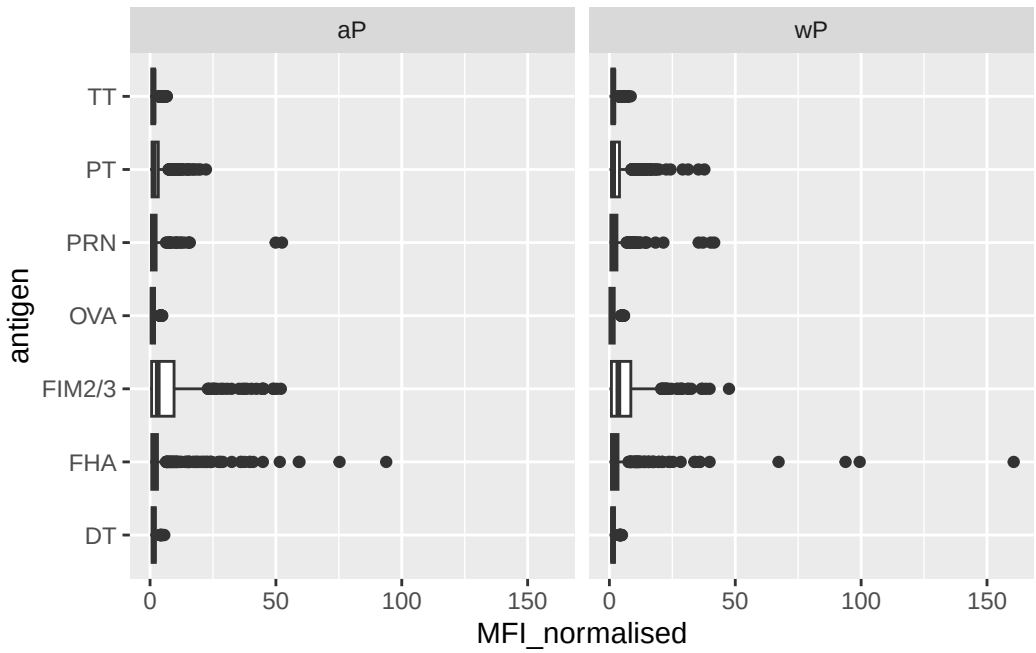
```
3                                -3                                0          Blood
4                                -3                                0          Blood
5                                -3                                0          Blood
6                                -3                                0          Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot()
```

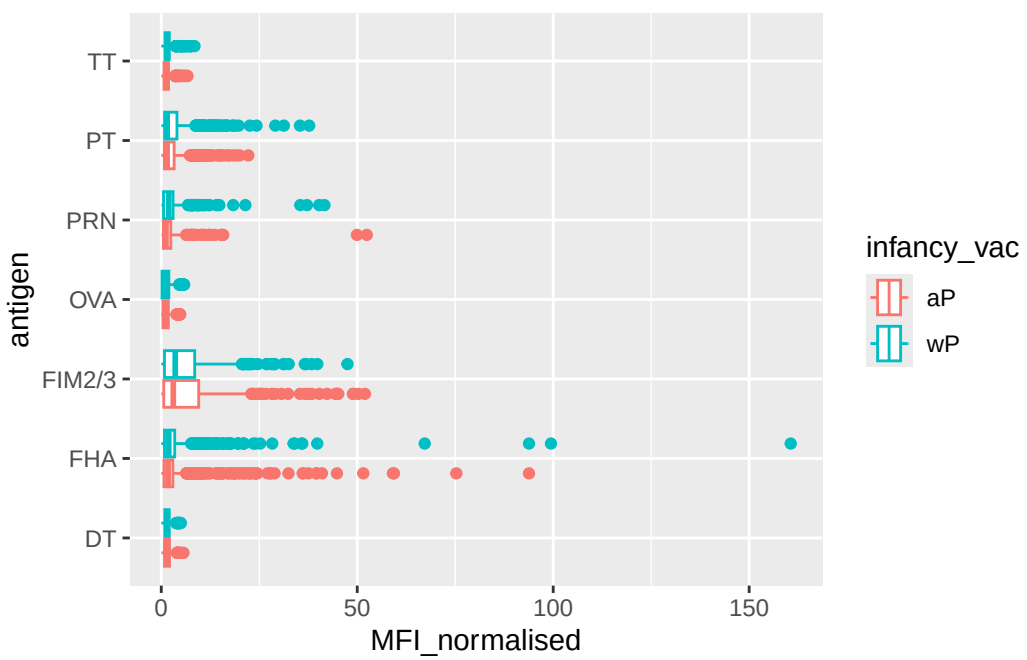


Q. Is there a difference for aP vs wP individuals with these values?

```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```

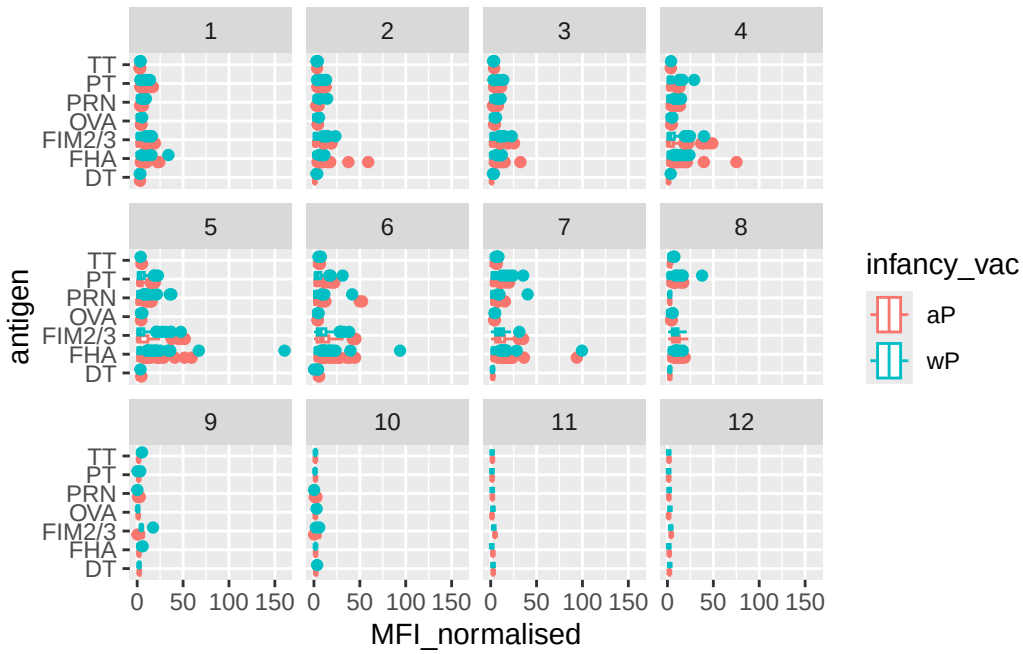



```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```



Q. Is there a temprol response - i.e. do values increase or decrease over time?

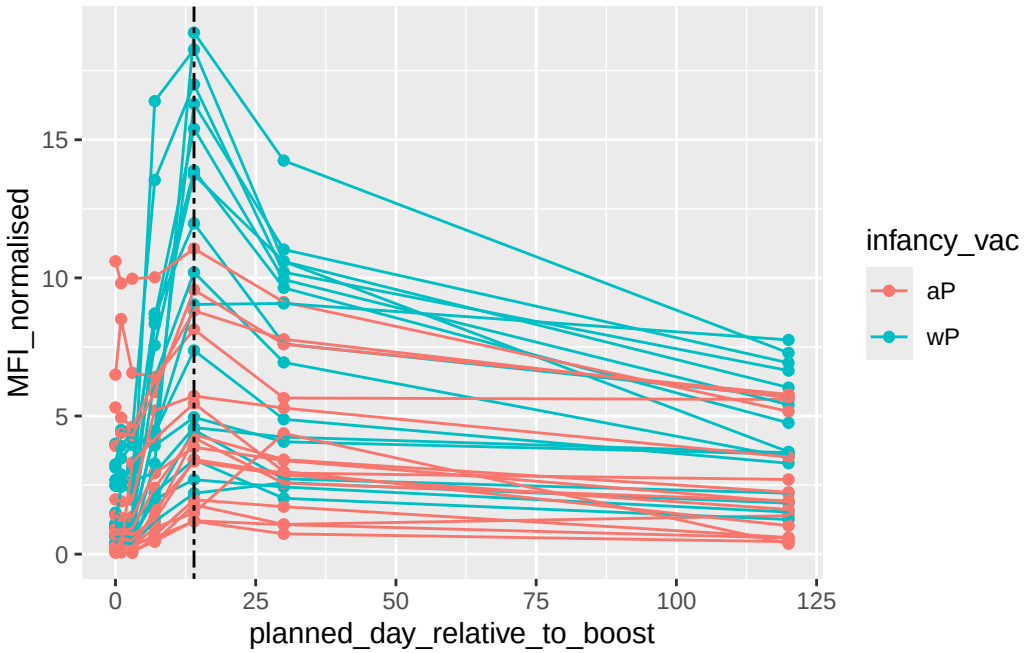
```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~visit)
```



FOcus on "PT" Pertussis TOxin antigen

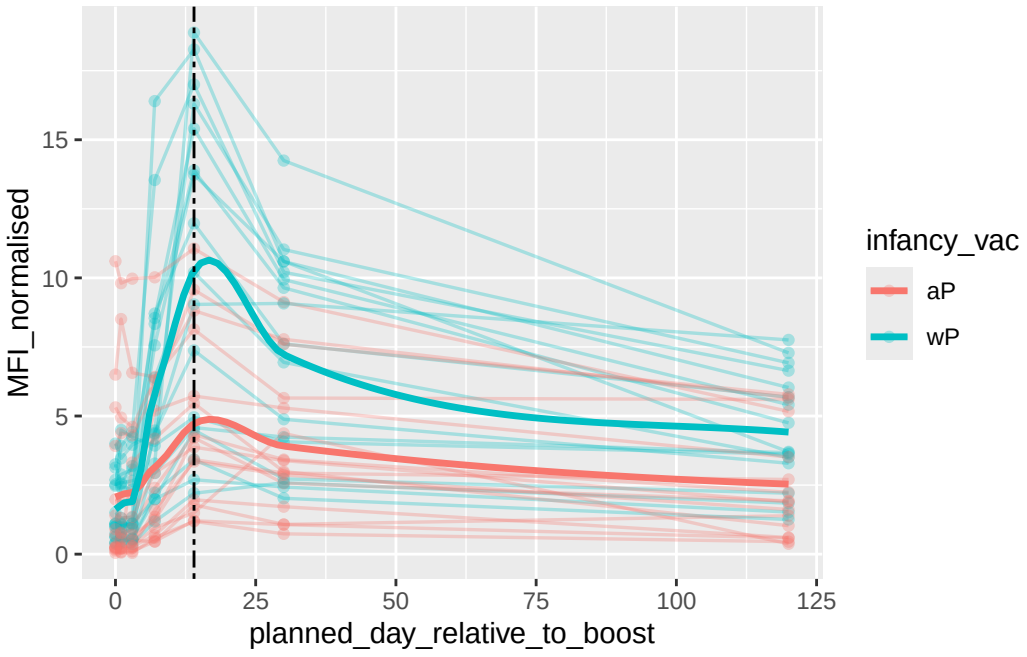
```
pt.igg.21 <- igg |> filter(antigen == "PT",
               dataset == "2021_dataset")
```

```
ggplot(pt.igg.21)+
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col=infancy_vac,
      group=subject_id)+
  geom_point()+
  geom_line()+
    geom_vline(xintercept = 14, lty=12)
```




```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col = infancy_vac) +


  geom_line(aes(group = subject_id),
            alpha = 0.3,
            linewidth = 0.6) +

  geom_point(alpha = 0.3) +


  geom_smooth(method = "loess",
              span=0.4,
              se = FALSE,
              linewidth = 1.2,
              alpha = 1) +

  geom_vline(xintercept = 14, lty = 12)
```

```
`geom_smooth()` using formula = 'y ~ x'
```


MFI_normalised
15
10
5
0
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP